

231501141
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Exercise 3: Google App Engine SDK and Web App Development

Aim

To install the Google App Engine SDK and develop a Hello World / simple web application using Python.

Procedure

Step 1: Install Google Cloud CLI & Python

Ensure Python 3 is installed on the system.

```
ubuntu@ubuntu:~$ sudo apt install python3
python3 is already the newest version (3.14.3-0ubuntu2).
python3 set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 3
```

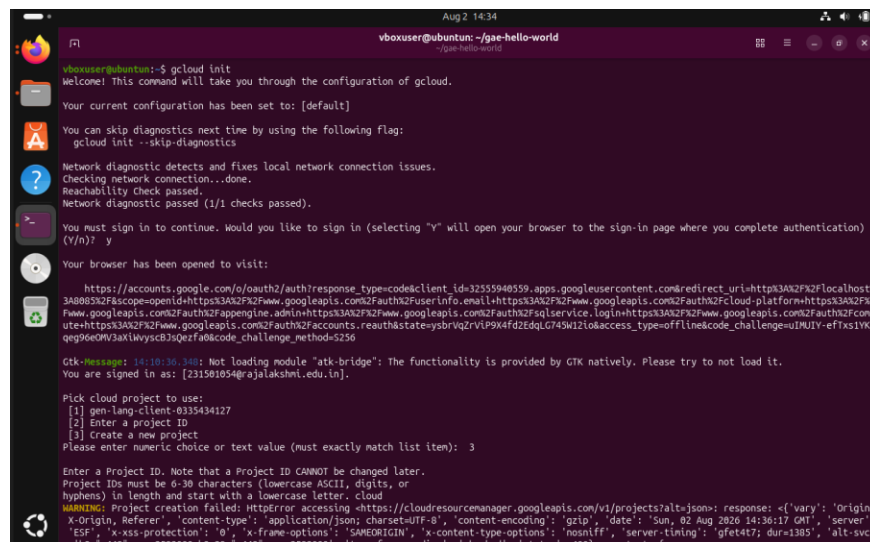
Download and install the Google Cloud CLI (contains the modern App Engine SDK) from cloud.google.com/sdk/docs/install:

```
curl -O https://dl.google.com/dl/cloudsdk/channels/rapid/downloads/google-cloud-cli-linux-x86_64.tar.gz
tar -xvf google-cloud-cli-linux-x86_64.tar.gz
./google-cloud-sdk/install.sh
```

Step 2: Initialize GCloud & Install Components

gcloud init

Sign in to your Google account when prompted.



```
Aug 2 14:34
vboxuser@ubuntu: ~/gae-hello-world
vboxuser@ubuntu:~$ gcloud init
Welcome! This command will take you through the configuration of gcloud.

Your current configuration has been set to: [default]

You can skip diagnostics next time by using the following flag:
gcloud init --skip-diagnostics

Network diagnostic detects and fixes local network connection issues.
Checking network connection...done.
Reachability check passed.
Network diagnostic passed (1/1 checks passed).

You must sign in to continue. Would you like to sign in (selecting "Y" will open your browser to the sign-in page where you complete authentication)
(Y/n): y

Your browser has been opened to visit:
https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555940559.apps.googleusercontent.com&redirect_uri=https%3A%2F%2Flocalhost%3A8080%2F&scope=openid%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fappengine.admin%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fsqlservice.login%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcompute%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Faccounts.reauth%2Fstate=ysbrvqzrVLP9X4fdEdqLG745K12Io&access_type=offline&code_challenge=UJUIV-efTxs1Ykyqeg96e0M73ak1WyscbJ3QezfAb&code_challenge_method=S256

Gtk-Message: 14:10:36.340: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
You are signed in as: [231501054@rajalakshmi.edu.in].

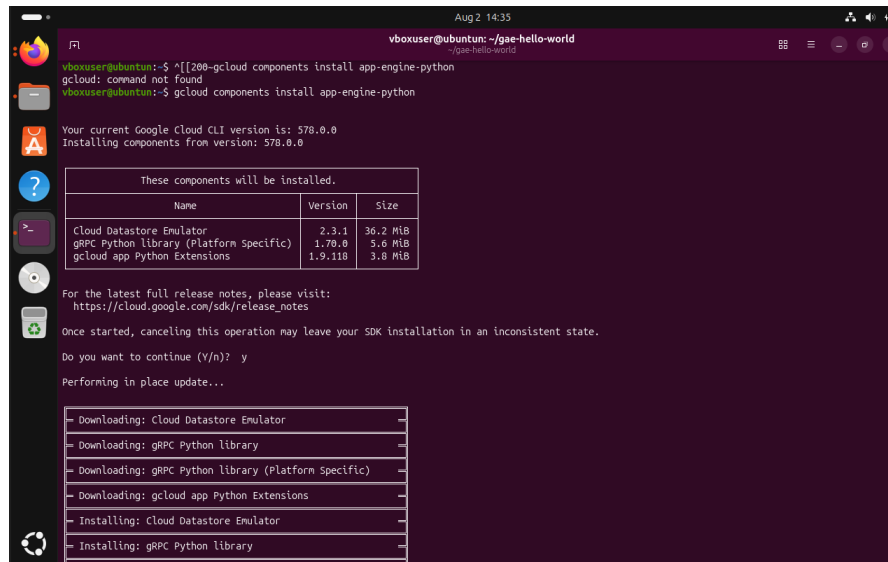
Pick cloud project to use:
[1] gen-lang-client-0335434127
[2] Enter a project ID
[3] create a new project
Please enter numeric choice or text value (must exactly match list item): 3

Enter a Project ID. Note that a Project ID CANNOT be changed later.
Project IDs must be 6-30 characters (lowercase ASCII, digits, or hyphens) in length and start with a lowercase letter. cloud
WARNING: Project creation failed: HttpError accessing <https://cloudresourcemanager.googleapis.com/v1/projects?alt=json>: response: <{'vary': 'Origin, X-Origin, Referer', 'content-type': 'application/json; charset=UTF-8', 'content-encoding': 'gzip', 'date': 'Sun, 02 Aug 2020 14:36:17 GMT', 'server': 'ESF', 'x-xss-protection': '0', 'x-frame-options': 'SAMEORIGIN', 'x-content-type-options': 'nosniff', 'server-timing': 'gfe47; dur=1305', 'alt-svc': '<h3>=443'; max-age=604800 h3-20='443'; max-age=604800'}> transfer-encoding: chunked; status: 400>. content of
```

231501141

Sailesh Rangaraj

gcloud components install app-engine-python



The screenshot shows a terminal window titled 'vboxuser@ubuntu: ~/gae-hello-world' with the date 'Aug 2 14:35'. The user enters the command 'gcloud components install app-engine-python'. The terminal output shows the current Google Cloud CLI version is 578.0.0 and that components are being installed from version 578.0.0. A table lists the components to be installed: Cloud Datastore Emulator (2.3.1, 36.2 MiB), gRPC Python library (Platform Specific) (1.70.0, 5.6 MiB), and gcloud app Python Extensions (1.9.118, 3.8 MiB). Below the table, a message states: 'For the latest full release notes, please visit: https://cloud.google.com/sdk/release_notes'. A confirmation prompt 'Do you want to continue (Y/n)? y' is shown, followed by 'Performing in place update...'. A progress bar shows the download and installation status for each component.

Name	Version	Size
Cloud Datastore Emulator	2.3.1	36.2 MiB
gRPC Python library (Platform Specific)	1.70.0	5.6 MiB
gcloud app Python Extensions	1.9.118	3.8 MiB

For the latest full release notes, please visit:
https://cloud.google.com/sdk/release_notes

Once started, canceling this operation may leave your SDK installation in an inconsistent state.

Do you want to continue (Y/n)? y

Performing in place update...

Downloading: Cloud Datastore Emulator	
Downloading: gRPC Python library	
Downloading: gRPC Python library (Platform Specific)	
Downloading: gcloud app Python Extensions	
Installing: Cloud Datastore Emulator	
Installing: gRPC Python library	

Step 3: Create the App Directory

```
mkdir gae-hello-world
```

```
cd gae-hello-world
```

Step 4: Create Application Code (main.py)

```
from flask import Flask
```

```
app = Flask(__name__)
```

```
@app.route("/")
```

```
def hello():
```

```
    return "Hello World from Google App Engine!"
```

```
if __name__ == "__main__":
```

```
    app.run(host="127.0.0.1", port=8080, debug=True)
```

Step 5: Create Configuration File (app.yaml)

```
runtime: python39
```

```
handlers:
```

```
- url: /*
```

```
  script: auto
```

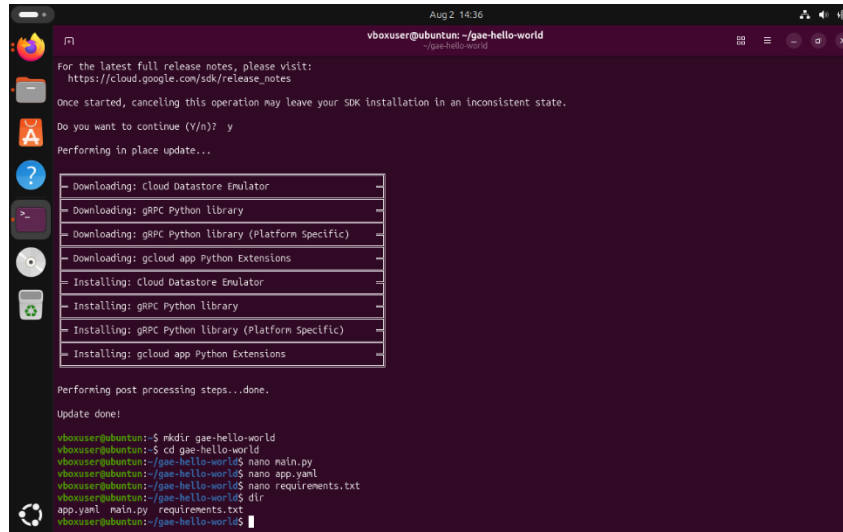
Step 6: Create requirements.txt

```
Flask==2.0.1
```

Output

Files created in project directory (main.py, app.yaml, requirements.txt):

231501141
Sailesh Rangaraj

A terminal window titled 'vboxuser@ubuntu: ~/gae-hello-world' showing the installation of the Google App Engine SDK. The terminal displays instructions to visit the release notes, a confirmation to continue, and a progress bar for downloading and installing various components. The components listed are: Cloud Datastore Emulator, gRPC Python library, gRPC Python library (Platform Specific), and gcloud app Python Extensions. The installation is shown as complete, followed by post-processing steps and a directory listing of the installed files.

```
Aug 2 14:36
vboxuser@ubuntu: ~/gae-hello-world

For the latest full release notes, please visit:
https://cloud.google.com/sdk/release_notes

Once started, cancelling this operation may leave your SDK installation in an inconsistent state.
Do you want to continue (Y/n)? y
Performing in place update...

- Downloading: Cloud Datastore Emulator
- Downloading: gRPC Python library
- Downloading: gRPC Python library (Platform Specific)
- Downloading: gcloud app Python Extensions
- Installing: Cloud Datastore Emulator
- Installing: gRPC Python library
- Installing: gRPC Python library (Platform Specific)
- Installing: gcloud app Python Extensions

Performing post processing steps...done.
Update done!

vboxuser@ubuntu: $ mkdir gae-hello-world
vboxuser@ubuntu: $ cd gae-hello-world
vboxuser@ubuntu: ~/gae-hello-world$ nano main.py
vboxuser@ubuntu: ~/gae-hello-world$ nano app.yaml
vboxuser@ubuntu: ~/gae-hello-world$ nano requirements.txt
vboxuser@ubuntu: ~/gae-hello-world$ dir
app.yaml  main.py  requirements.txt
vboxuser@ubuntu: ~/gae-hello-world$
```

Result

Thus, the Google App Engine SDK was installed and a simple Hello World web application was developed using Python and Flask.